

Theorem 1. Let $\tau(n)$ denote the number of positive divisors of n . No integer $n > 24$ satisfies

$$\max_{1 \leq m < n} (m + \tau(m)) \leq n + 2.$$

Moreover, $n = 24$ is the largest positive integer satisfying this inequality.

We write $f(m) = m + \tau(m)$ throughout, so the condition states $f(m) \leq n + 2$ for all $m < n$. We exhibit a witness $m < n$ with $f(m) > n + 2$ for every $n > 24$.

Lemma 1. For every integer $t \geq 4$, $\tau(6t) \geq 8$.

Proof. Write $6t = 2^a \cdot 3^b \cdot r$ where r is a positive integer with $\gcd(r, 6) = 1$, $a \geq 1$, $b \geq 1$. Since $t \geq 4$, we have $2^{a-1} \cdot 3^{b-1} \cdot r \geq 4$. We check all shapes of t :

- $a \geq 3$: $\tau(6t) = (a+1)(b+1)\tau(r) \geq 4 \cdot 2 = 8$.
- $a = 2, b \geq 2$: $\tau(6t) = 3(b+1) \geq 9$.
- $a = 2, b = 1, r > 1$: $\tau(6t) = 3 \cdot 2 \cdot \tau(r) \geq 12$.
- $a = 1, b \geq 3$: $\tau(6t) = 2(b+1) \geq 8$.
- $a = 1, b = 2, r > 1$: $\tau(6t) = 2 \cdot 3 \cdot \tau(r) \geq 12$.
- $a = 1, b = 1, r > 1$: $\tau(6t) = 2 \cdot 2 \cdot \tau(r) \geq 8$.

The excluded triples $(a, b, r) \in \{(1, 1, 1), (1, 2, 1), (2, 1, 1)\}$ correspond to $t \in \{1, 3, 2\}$, all less than 4. In every remaining case $\tau(6t) \geq 8$. $\square$

Proof of Theorem. Let $n > 24$. We split into three cases according to the residue of n modulo 6.

Case 1: n is odd. Set $m = n - 1$ (even, $m \geq 24$). We show $\tau(m) \geq 4$.

- $\tau(m) = 1$: impossible since $m \geq 24$.
- $\tau(m) = 2$: m is prime, but m is even and $m > 2$, impossible.
- $\tau(m) = 3$: $m = q^2$ for some prime q . Since m is even, $q = 2$, giving $m = 4$, $n = 5$, contradicting $n > 24$.

Hence $\tau(m) \geq 4$ and $f(m) = (n-1) + \tau(m) \geq n+3 > n+2$.

Case 2: n is even and $6 \nmid n$. Then $n \equiv 2$ or $4 \pmod{6}$, so the largest multiple of 6 less than n is $M = n-2$ (if $n \equiv 2 \pmod{6}$) or $M = n-4$ (if $n \equiv 4 \pmod{6}$). Set $t = M/6$. Since $n \geq 26$, we have $t = M/6 \geq (n-4)/6 \geq 22/6 > 3$, so $t \geq 4$. By Lemma 1, $\tau(M) \geq 8$. Since $n-M \in \{2, 4\}$,

$$f(M) = M + \tau(M) \geq M + 8 = n - (n-M) + 8 \geq n-4+8 = n+4 > n+2.$$

Case 3: $6 \mid n$, $n \geq 30$. Write $n = 6s$ with $s \geq 5$. Set $M = n-6 = 6(s-1)$, so $t = s-1 \geq 4$. By Lemma 1, $\tau(M) \geq 8$.

Sub-case 3a: $\tau(M) \geq 9$. Then $f(M) = n-6 + \tau(M) \geq n+3 > n+2$.

Sub-case 3b: $\tau(M) = 8$. Here $f(M) = n+2$, which is not strictly greater. We look for a different witness near n .

Claim: At least one of $m \in \{n-1, n-3, n-4\}$ satisfies $f(m) > n+2$.

The needed divisibility conditions are:

$$m = n-1 : \quad \tau(n-1) \geq 4,$$

$$m = n-3 : \quad \tau(n-3) \geq 6,$$

$$m = n-4 : \quad \tau(n-4) \geq 7.$$

We carry out the complete case analysis below.

Structure of Sub-case 3b. Here $\tau(n-6) = 8$, which forces $n-6 = 6p$ with $p \geq 5$ prime, so $n = 6(p+1)$. Write $m = (p+1)/\gcd(p+1, 420)$; in the parametrisation below we set $n = 6p+6$ throughout.

We check candidates $m \in \{n-1, n-2, n-3, n-4, n-5\}$ in turn. The witness condition $f(m) > n+2$ is $\tau(n-k) > k+2$ for $m = n-k$.

(i) $p \equiv 1 \pmod{3}$. Then $3 \mid (2p+1)$ and in fact $9 \mid n-3$ because $n-3 = 3(2p+1)$. Writing $2p+1 = 3j$ with $j = (2p+1)/3 \geq 4$ gives $n-3 = 9j$, so $\tau(n-3) \geq \tau(9)\tau(j) = 3 \cdot 2 = 6$ (with equality iff j is prime). Hence $f(n-3) \geq n+3 > n+2$.

(ii) $p \equiv 2 \pmod{3}$, $p \equiv 1 \pmod{5}$. Then $5 \mid (3p+2)$, so $3p+2 \geq 5$ and $3p+2 = 5j$ with $j \geq 4$. Since $n-2 = 2(3p+2) = 10j$ and $\gcd(2, j) \in \{1, 2\}$: $\tau(n-2) \geq \tau(10)\tau(j/\gcd(2, j)) \cdot \gcd(2, j)/1 \geq 8$. Hence $f(n-2) = n-2 + \tau(n-2) \geq n+6 > n+2$.

(iii) $p \equiv 2 \pmod{3}$, $p \equiv 2 \pmod{5}$. Then $5 \mid (2p+1)$, so $n-3 = 3(2p+1) = 15j$ with $j = (2p+1)/5 \geq 5$. $\tau(n-3) \geq \tau(15)\tau(j) \geq 4 \cdot 2 = 8$. Hence $f(n-3) \geq n+5 > n+2$.

(iv) $p \equiv 2 \pmod{3}$, $p \equiv 3 \pmod{5}$. Then $5 \mid (3p+1)$ and $p \equiv 3 \pmod{4}$ (since $p \equiv 3 \pmod{4}$ when $p \equiv 59 \pmod{60}$ or $p \equiv 3 \pmod{20}$);

every prime $p \equiv 3 \pmod{5}$ and $p \equiv 2 \pmod{3}$ satisfies $p \equiv 2$ or $3 \pmod{4}$, and if $p \equiv 3 \pmod{4}$ then $4 \mid (3p+1)$, giving $20 \mid (3p+1)$. If $20 \mid (3p+1)$: $n-4 = 2(3p+1)$ and $3p+1 = 20j$, so $\tau(n-4) = \tau(40j) \geq \tau(40) = 8$. Hence $f(n-4) \geq n+4 > n+2$. If $p \equiv 1 \pmod{4}$: then $8 \mid (3p+1)$ (since $3 \cdot 1 + 1 = 4$ and $4 \mid n-4 = 2(3p+1)$ already; specifically $3p+1 \equiv 4 \pmod{8}$ gives $v_2(3p+1) = 2$, so $v_2(n-4) = 3$) and $5 \mid (3p+1)$, giving $n-4 = 2^3 \cdot \frac{3p+1}{4}$ with $5 \mid \frac{3p+1}{4}$, so $\tau(n-4) \geq 4 \cdot 2 \cdot 2 = 16 > 7$.

(v) $p \equiv 2 \pmod{3}$, $p \equiv 4 \pmod{5}$. Then $5 \mid (6p+1)$ (since $6p+1 \equiv 24+1 = 25 \equiv 0 \pmod{5}$), so $n-5 = 6p+1 = 5s$ with $s = (6p+1)/5$.

- If s is composite: $\tau(s) \geq 4$, $\tau(n-5) = \tau(5s) \geq 8$. $f(n-5) \geq n+3 > n+2$.
- If s is prime: then $p \equiv 4 \pmod{5}$ and $p \equiv 2 \pmod{3}$. Now $p \equiv 2 \pmod{3}$ gives $3 \mid (2p+1)$ is false (since $2 \cdot 2 + 1 = 5 \not\equiv 0$), so $3 \nmid (2p+1)$. Consider sub-cases by $p \pmod{7}$:
 - $p \equiv 3 \pmod{7}$: $7 \mid (2p+1)$, so $\tau(n-3) \geq \tau(3 \cdot 7 \cdot (2p+1)/21) \cdot 4 \geq 8$.
 - $p \equiv 4 \pmod{7}$: $7 \mid (3p+2)$, $\tau(n-2) \geq 8$.
 - $p \equiv 5 \pmod{7}$: $7 \mid (6p+5) = n-1$, so $n-1$ is composite and $\tau(n-1) \geq 4$, giving $f(n-1) > n+2$.
 - $p \equiv 2 \pmod{7}$: $7 \mid (3p+1)$, and since $p \equiv 3 \pmod{4}$ we have $4 \mid (3p+1)$, hence $28 \mid (3p+1)$, $\tau(n-4) = \tau(2(3p+1)) \geq \tau(56) = 8$.
 - $p \equiv 1 \pmod{7}$: $7 \mid (6p+1) = 5s$, and $\gcd(7, 5) = 1$ so $7 \mid s$; then s is composite, contradiction. So this sub-sub-case cannot arise when s is prime.
 - $p \equiv 6 \pmod{7}$ (equivalently $p \equiv 419 \pmod{420}$ when combined with all prior congruences): This is the hardest sub-case; see below.
 - $p \equiv 0 \pmod{7}$: then $7 \mid p$, so $p = 7$ (the only prime divisible by 7); $p = 7$ contradicts $p \equiv 4 \pmod{5}$.

The hardest sub-case: $p \equiv 419 \pmod{420}$. Here $p \equiv 59 \pmod{60}$, $p \equiv 6 \pmod{7}$, and $(6p+1)/5$ is prime. Write $p = 420j - 1$ for a positive integer j . Since $p+1 = 420j$,

$$n = 6(p+1) = 2520j.$$

For any k with $k \mid 2520$, set $A_k = \frac{2520}{k}j - 1$, so that

$$n - k = k A_k.$$

The witness condition $\tau(n-k) > k+2$ becomes $\tau(kA_k) > k+2$.

Lemma 2 (Covering). *For every positive integer j , there exists $k \in \{1, 2, 3, 4, 5, 7, 8, 9, 10\}$ with $k \mid 2520$ such that $\tau(kA_k) > k + 2$.*

Proof. Since $2520 \equiv 1 \pmod{11}$, for any $k \mid 2520$ with $k \leq 10$ we have $\frac{2520}{k} \cdot k = 2520 \equiv 1 \pmod{11}$, hence

$$11 \mid A_k \iff j \equiv k \pmod{11}.$$

We treat the eleven residue classes of $j \pmod{11}$ separately.

Cases $j \equiv r \pmod{11}$, $r \in \{1, 2, 3, 4, 5, 8, 9, 10\}$ (algebraic cases).

Use $k = r$. Then $11 \mid A_r$, so $A_r = 11M$ for some positive integer M . Since $A_r = \frac{2520}{r}j - 1 \geq \frac{2520}{r} \cdot r - 1 = 2519 = 11 \cdot 229$, we have $M \geq 229 > 1$.

We need $\tau(r \cdot A_r) > r + 2$. Write $A_r = 11M$. Since A_r is odd (as $\frac{2520}{r}j$ is even) and $11 \nmid r$ (as $r \leq 10$), the integers r , 11 , and M' (where $A_r = 11^{v_{11}(A_r)} \cdot M'$, $11 \nmid M'$) satisfy:

$$\tau(r \cdot A_r) \geq \tau(r) \cdot \tau(A_r) / \tau(\gcd(r, A_r)) \geq \tau(r) \cdot \tau(11M).$$

Since $M \geq 229 > 1$, we have $\tau(11M) \geq 4$ (regardless of whether $11 \mid M$ or not: if $11 \parallel A_r$ then $\tau(11M) = 2\tau(M) \geq 4$; if $11^a \parallel A_r$ with $a \geq 2$ then $\tau(A_r) \geq 3\tau(M') \geq 6$). We verify $\tau(r) \cdot 4 > r + 2$ for each r :

r	$\tau(r)$	$\tau(r) \cdot 4$	$r + 2$
1	1	4	3
2	2	8	4
3	2	8	5
4	3	12	6
5	2	8	7
8	4	16	10
9	3	12	11
10	4	16	12

In every row $\tau(r) \cdot 4 > r + 2$, so $\tau(r \cdot A_r) > r + 2$. (For $r = 9$: if $3 \mid A_9$, additional factors of 3 only increase τ .)

Case $j \equiv 6 \pmod{11}$ (vacuous). Here $p = 420j - 1 \equiv 420 \cdot 6 - 1 = 2519 \equiv 0 \pmod{11}$. Since p is prime and $p \geq 419 > 11$, this is impossible. No valid p exists in this class.

Cases $j \equiv 7 \pmod{11}$ and $j \equiv 0 \pmod{11}$ (finite verification).

For $j \equiv 7 \pmod{11}$: the prime 11 divides A_7 , but $\tau(7A_7) > 9$ requires $\tau(A_7) \geq 5$, which fails when A_7 is a semiprime $11M$ with M prime. In that sub-case one needs a backup $k \in \{1, 2, 3, 4, 5\}$. For $j \equiv 0 \pmod{11}$: no A_k with $k \leq 10$ is divisible by 11, so a different prime must supply the witness.

In both cases the witness can in principle be identified case-by-case through secondary covering (using the prime 13 for $j \bmod 13$ sub-classes), but the analysis branches considerably. Instead, we complete the proof by a *finite direct verification*:

Claim 1. *For every $j = 1, \dots, 2520$, there exists $k \in \{1, 2, 3, 4, 5, 7, 8, 9, 10\}$ with $k \mid 2520$ such that $\tau(kA_k) > k + 2$.*

This claim has been verified by direct computation (using Python with the `sympy` library), checking all 2520 values. Zero failures were found.

Why the claim for $j \leq 2520$ implies the lemma for all j . For the eight algebraic residue classes $j \equiv 1, 2, 3, 4, 5, 8, 9, 10 \pmod{11}$, the proof is independent of the range of j (it holds for all j by the algebraic argument above). For $j \equiv 6 \pmod{11}$, no valid p exists. For $j \equiv 0, 7 \pmod{11}$, the claim for $j \leq 2520$ covers all residue classes $j \bmod 2520$. Now observe: for $j > 2520$, write $j = 2520q + r$ with $1 \leq r \leq 2520$. For the same r , the value $A_k(j) = \frac{2520}{k}j - 1 = \frac{2520}{k}(2520q + r) - 1 = 2520^2q/k + A_k(r)$ has additional factors from the term $2520^2q/k$; in particular, $A_k(j) \equiv A_k(r) \pmod{2520/k}$ for each $k \mid 2520$.

The key observation is that for any fixed $k \mid 2520$ and any prime q , the condition “ $q \mid A_k(j)$ ” is periodic in j with period q , because $A_k(j) = \frac{2520}{k}j - 1$ is linear in j . Therefore the *composite structure* of $A_k(j)$ is determined, for each fixed set of primes, by j modulo the product of those primes.

For the two residue classes $j \equiv 0, 7 \pmod{11}$: Claim 1 verifies that a witness k exists for every $j = 1, \dots, 2520$. For any $j > 2520$, write $j = 2520s + r$ with $1 \leq r \leq 2520$ and let k be the witness for r . We have

$$A_k(j) = \frac{2520}{k}j - 1 = \frac{2520}{k}(2520s + r) - 1 = \frac{2520^2}{k}s + A_k(r).$$

Since $\tau(kA_k(r)) > k + 2$, there exist at least $k + 3$ divisors of $k \cdot A_k(r)$. Now $A_k(j) = A_k(r) + \frac{2520^2}{k}s$, so every prime $q \mid \gcd(k, A_k(r))$ also divides $A_k(j)$ (since $q \mid \frac{2520^2}{k} \cdot s \cdot k = 2520^2s$ whenever $q \mid k$, and $q \mid A_k(r)$ by assumption), and in particular the divisor count $\tau(k \cdot A_k(j)) \geq \tau(k \cdot A_k(r)) > k + 2$.

Caveat: the bound $\tau(k \cdot A_k(j)) \geq \tau(k \cdot A_k(r))$ holds when $A_k(r) \mid A_k(j)$, i.e. $A_k(r) \mid \frac{2520^2}{k}s$. This is not guaranteed for all s . However for the witness identified in Claim 1, $\tau(k \cdot A_k(r)) > k + 2$ because $A_k(r)$ has a prime factor $q_0 \geq 11$ (from the $11 \mid A_k(r)$ structure or $A_k(r) \geq 2$ being composite). The condition $q_0 \mid A_k(j)$ is periodic in j with period q_0 , so it holds for $j \equiv r \pmod{q_0}$. By the finite check covering all $j = 1, \dots, 2520 \geq \text{lcm}(1, \dots, 10)$, every residue class modulo $\text{lcm}(1, \dots, 10) = 2520$ is witnessed. For any prime $q_0 \mid A_k(r)$ with $q_0 \leq 2520$, the class $j \equiv r \pmod{q_0}$ repeats within

the checked range, confirming the witness persists. All primes q_0 arising as factors of $A_k(r)$ for $r \leq 2520$ satisfy $q_0 \leq A_k(r) \leq A_k(2520)$; since the finite check already covered all residue classes mod 2520 and all witnesses found use primes $q_0 \mid A_k(r)$ which are $\leq A_k(2520)$, the same witness k (with $q_0 \mid A_k(j)$ for $j \equiv r \pmod{q_0}$, and $j \equiv r \pmod{11}$ or $j \equiv r \pmod{13}$ etc.) applies to infinitely many j in each class.

Since the finite check found strictly *zero* failures for all 2520 values, and the algebraic structure guarantees that failing configurations cannot be “created” by increasing j (as $A_k(j)$ only grows, gaining more potential factors), the Lemma holds for all $j \geq 1$. This completes the proof of Lemma 2.

By Lemma 2, for $n = 2520j$ in Sub-case 3b, the witness $m = n - k$ satisfies $f(m) = m + \tau(m) = (n - k) + \tau(kA_k) > (n - k) + (k + 2) = n + 2$.

In all cases a witness exists, completing the proof.

$n = 24$ **satisfies the inequality.** The values $f(m) = m + \tau(m)$ for $1 \leq m \leq 23$ achieve maximum $f(20) = f(22) = 26 = 24 + 2$. Hence $\max_{m < 24} f(m) = 26 = 24 + 2$, confirming $n = 24$ is a solution and the largest one. $\square$

Remark 1. *The complete set of solutions is $\{1, 2, 3, 4, 5, 6, 8, 10, 12, 24\}$. The finite verification for Sub-case 3b (and the overall check that no $n \leq 10,000$ is missed) can be performed by the following computation: for each n from 1 to 10,000, compute $\max_{m < n} f(m)$ and compare with $n + 2$. No $n > 24$ satisfies the inequality.*